Modernize your applications with purpose-built AWS databases

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Modern data architecture

From traditional . . .

. . . to microservices, decoupled architectures

Web servers

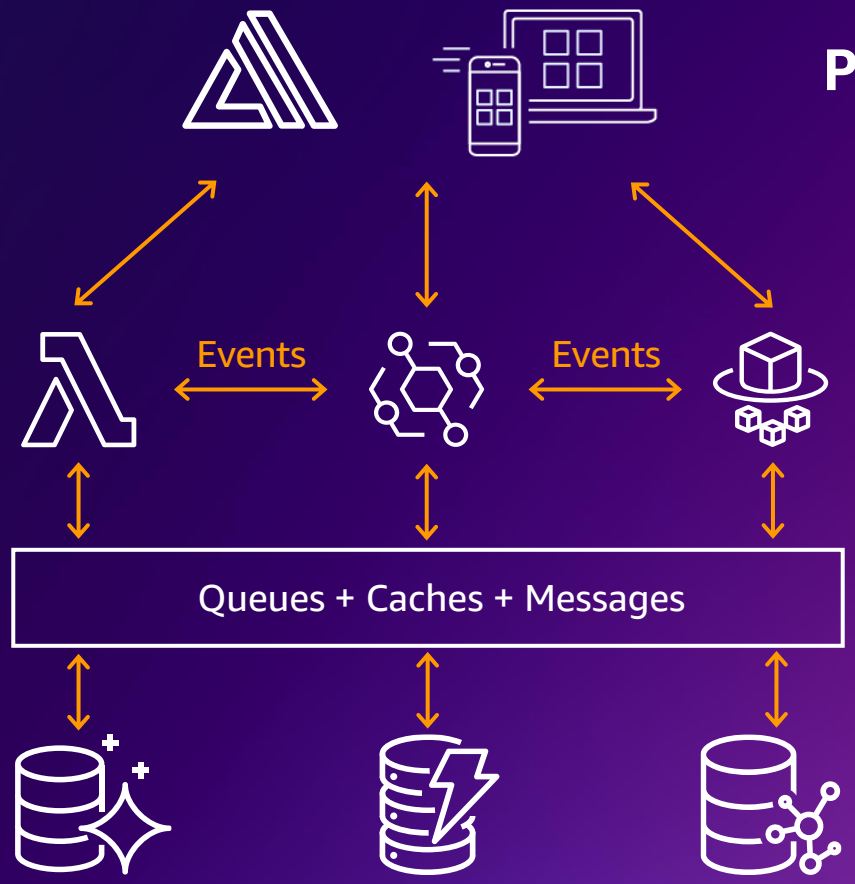
Presentation layers

Application servers

Business logic

Database servers

Data layer



Presentation

Business logic

Data

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The pursuit of the "ideal" database

THE IDEAL DATABASE SHOULD DO EVERYTHING WELL

This doesn't exist!

Innovation and agility

Supports **all storage** (data types) and **all access patterns**

Implements **ACID transactions** and **strong consistency**

Performance and scalability

Provides **limitless scalability** and **ultra low latency**

Is **highly performant** regardless of query complexity

Cost-effective and easy to manage

Is **continuously available** (no downtime)

Is **simple and easy to use, tune, and maintain**

Is **cost-effective** and **predictable**

. . . and more!

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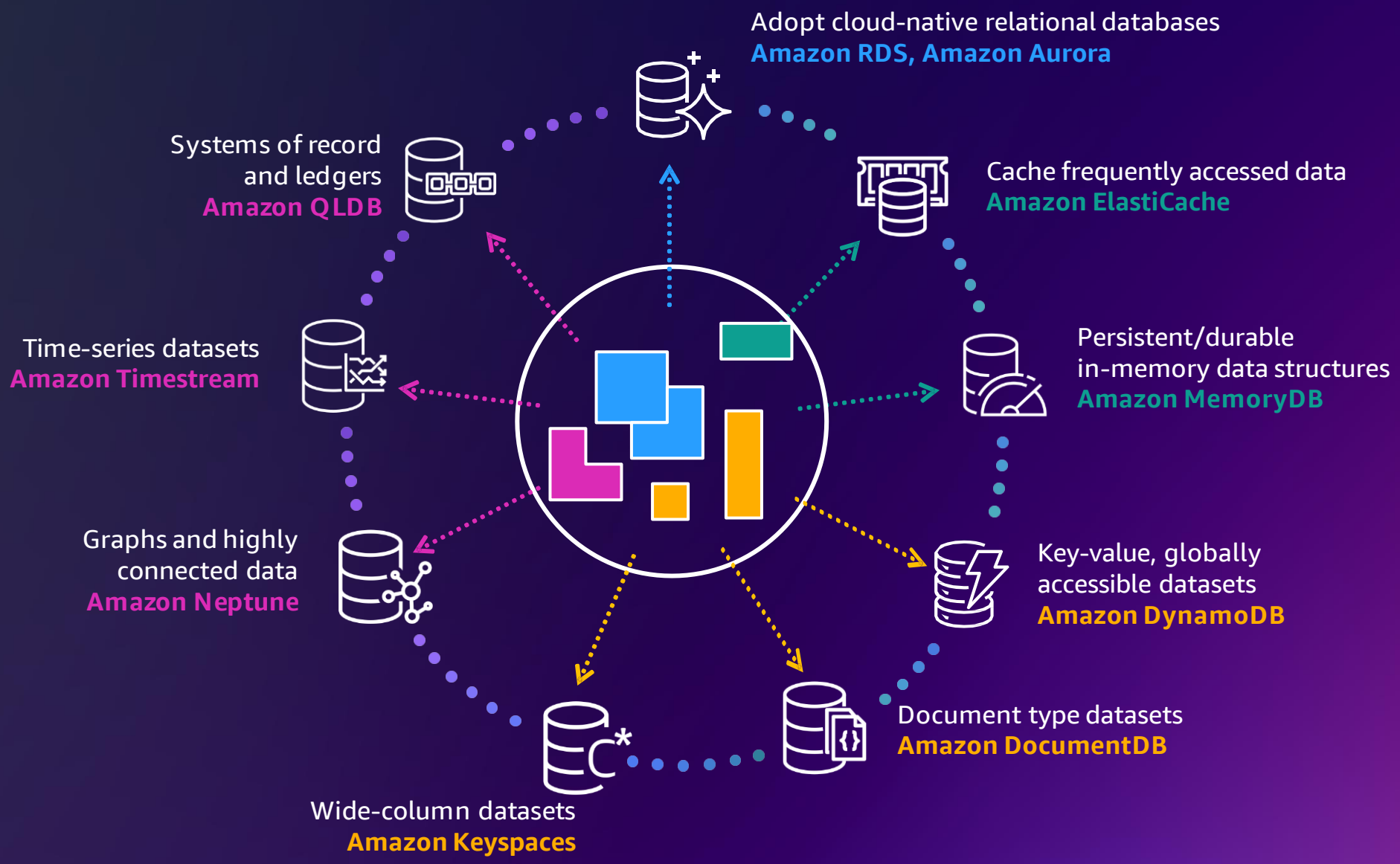
AWS purpose-built databases



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Modernization using purpose-built databases

It’s a journey!

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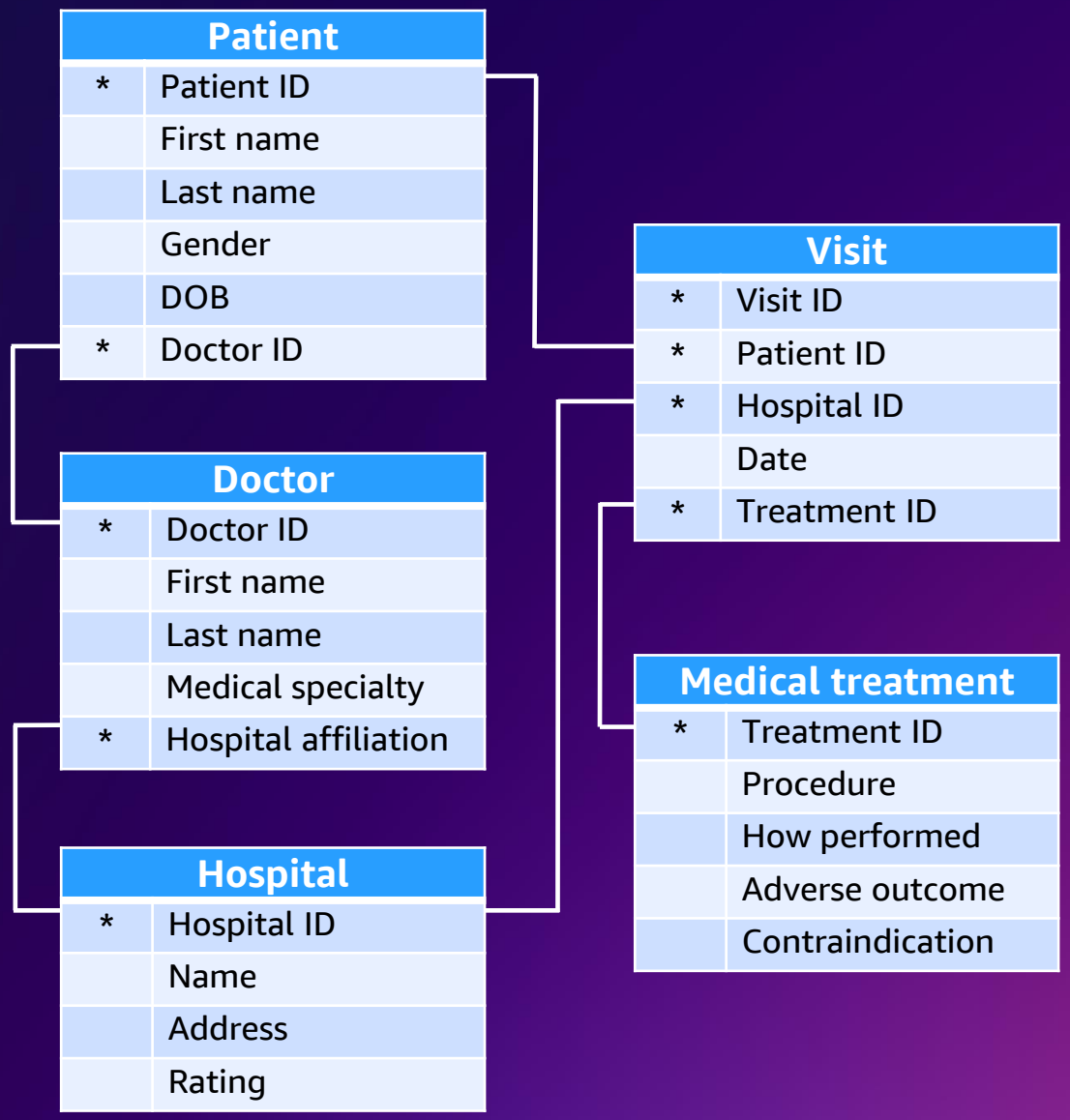
Relational data

Divide data among tables

Highly structured

Relationships established via keys enforced by the system

Data accuracy and consistency



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Step 1: Managed relational databases

Managed databases

- Reduce operational burden
- Take advantage of platform features

Benefits

- Agility
- Easily managed
- Cost-effective

Relational database application

Adopt cloud-native relational databases
Amazon RDS, Amazon Aurora

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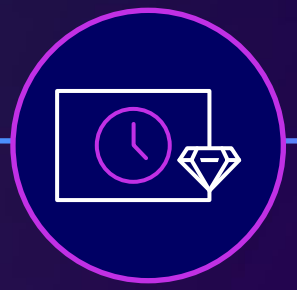
Amazon Relational Database Service (RDS)

Managed relational database service with a choice of seven popular database engines

Easy to administer

Easily deploy and maintain hardware, OS and DB software; built-in monitoring

Available and durable

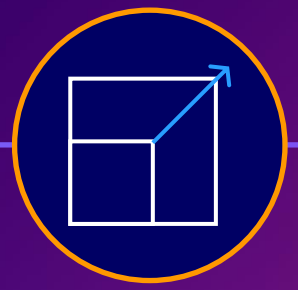


Automatic Multi-AZ data replication; automated backup, snapshots, and failover

Secure and compliant

Data encryption at rest and in transit; industry compliance and assurance programs

Highly scalable



Scale compute and storage with a few clicks; minimal downtime for your application

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Amazon Aurora

MySQL and PostgreSQL compatible relational database built for the cloud

Performance and scalability

Enterprise grade performance

Scale out
up to 15 read replicas

Availability and durability

Fault-tolerant, self-healing storage; six copies of data across three AZs; continuous backup to Amazon S3

Highly secure

Network isolation, encryption at rest / transit

Fully managed

Managed by RDS: No server provisioning, software patching, setup, configuration, or backups

aws

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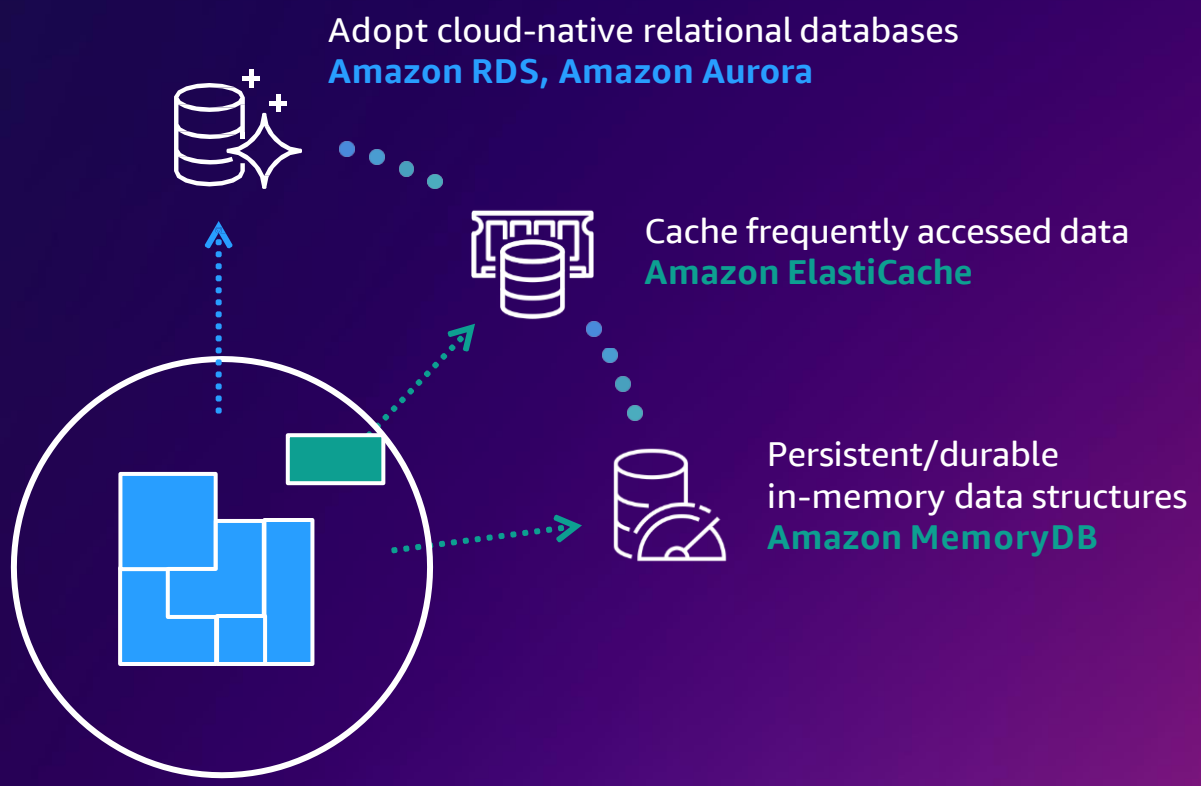
Step 2 : Cache and persistent in-memory databases

Address scaling pain

- Low-hanging fruit, easy to adopt
- Offload repetitive read operations
- Offload session state
- Caching

Increase performance

- Frequent counters
- Fast-changing rankings
- Submillisecond data access



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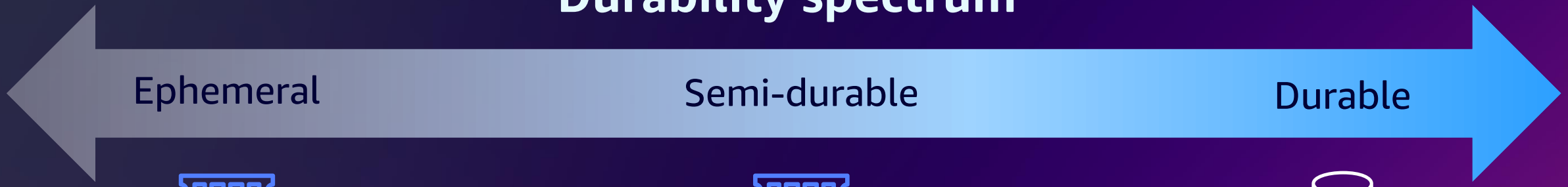
AWS in-memory managed solutions

Durability spectrum



Amazon ElastiCache for Memcached

Amazon ElastiCache for Redis

Amazon MemoryDB for Redis

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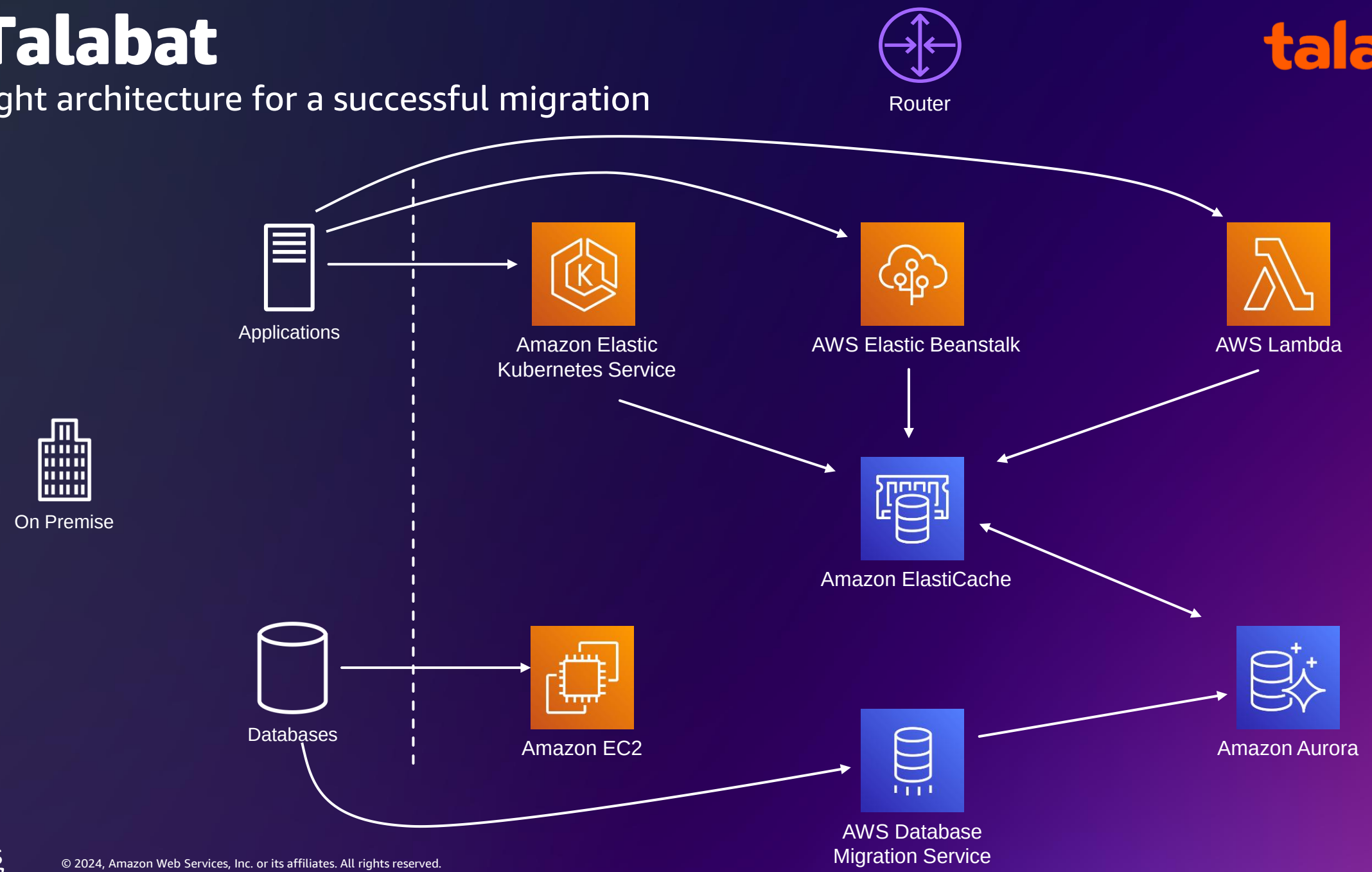
Talabat

Right architecture for a successful migration



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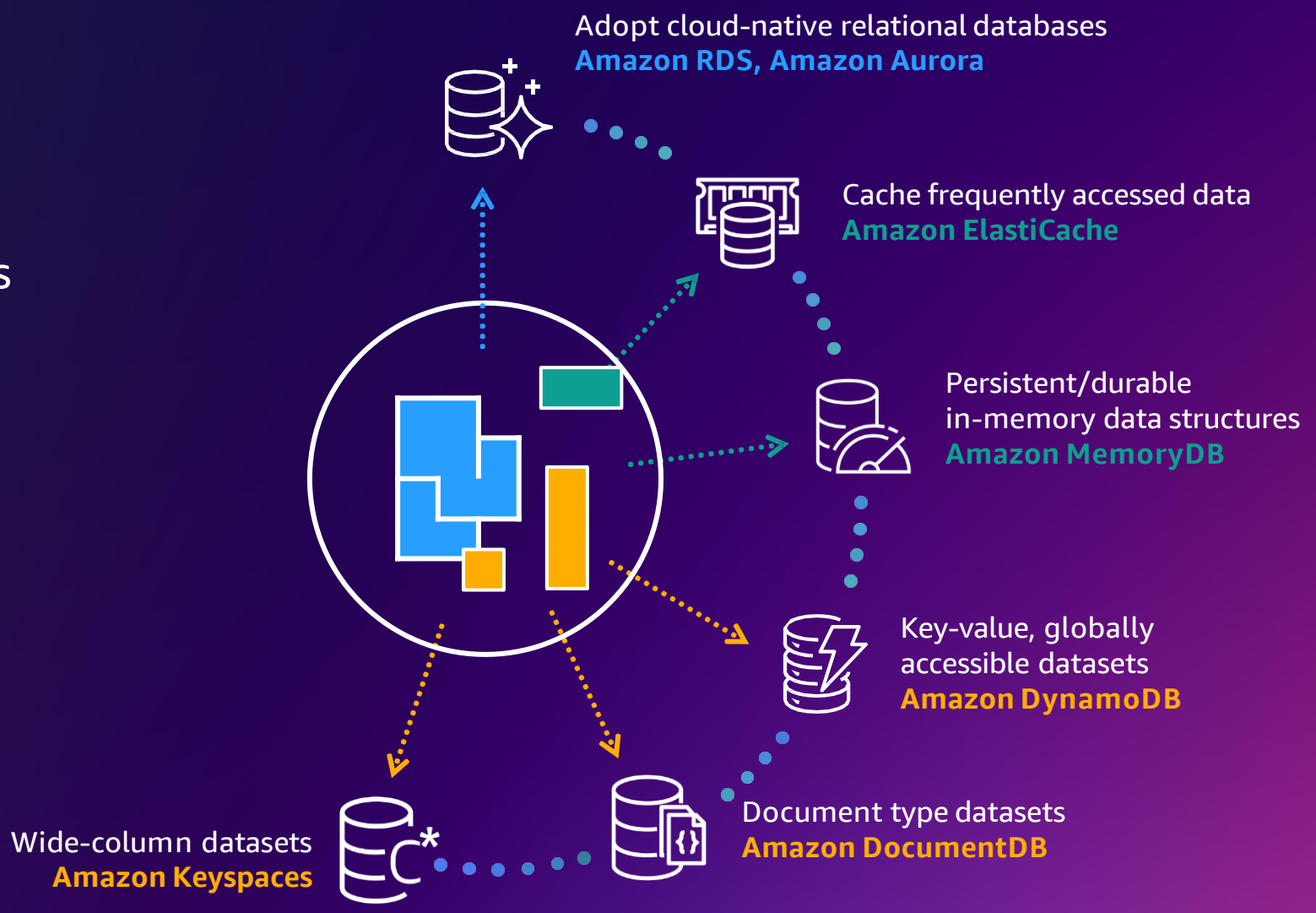
Step 3: Nonrelational access pattern databases

Breaking down the monolith

- Right tool for the job
- Microservice access pattern drives data store decision

Benefits

- Agility
- Scalability
- Performance



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Key-value data

Simple key-value pairs

Resilient to failure

High-throughput, low-latency reads and writes

Consistent performance at scale

```
PUT {
  TableName:"Gamers",
   Item: {
   "GamerTag":"Hammer57",
   "Level":21,
   "Points":4050,
   "Score":483610,
   "Plays":1722
    }
 }
```

```
GET {
  TableName:"Gamers",
  Key: {
    "GamerTag":"Hammer57“,
    “ProjectionExpression“:”Points”
    }
}
```

Gamers				
Primary key	Attributes			
Gamer tag	Level	Points	High score	Plays
Hammer57	21	4050	483610	1722
FluffyDuffy	5	1123	10863	43
Lol777313	14	3075	380500	1307
Jam22Jam	20	3986	478658	1694
ButterZZ_55	7	1530	12547	66
…	…	…	…	…

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Amazon DynamoDB

Fast and flexible key-value database service for any scale

Performance at scale

Consistent, single-digit millisecond response times at any scale; build applications with virtually unlimited throughput

Serverless architecture

No hardware provisioning, software patching or upgrades; scale up or down automatically; continuously back up your data

Enterprise security

Encrypt all data by default and fully integrate with AWS Identity and Access Management for robust security

Global replication

Build global applications with fast access to local data by easily replicating tables across multiple AWS Regions

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Novartis – Buying engine

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Why document databases?

- **Documents map naturally to how humans** model data
- Documents (objects / JSON) are common application data models
- Document databases provide flexible schema and indexing
- Ad hoc querying and aggregations

```
#Ad hoc queries
db.col.find( { "name.last": "Smith" } )

#Count of documents
db.col.find({ product: "xyz", region: "msp" }).count()

#Return distinct fields
db.col.distinct( "region" )

#Power aggregations over documents
db.col.aggregate([
                    {$match: {product: "xyz"}},
                    {$group: {_id: "$c_id", rev: {$sum: "$price"}}},
                    {$sort: {rev: -1 }}
                  ])
```

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Amazon DocumentDB

Fast, scalable, and highly available MongoDB-compatible database service

Fast and scalable

Millions of requests per second, millisecond latency

MongoDB compatible

Same code, drivers, and tools you use with MongoDB

Fully managed

Simple and fully managed; automated monitoring and alerting

Enterprise security

Secure and compliant

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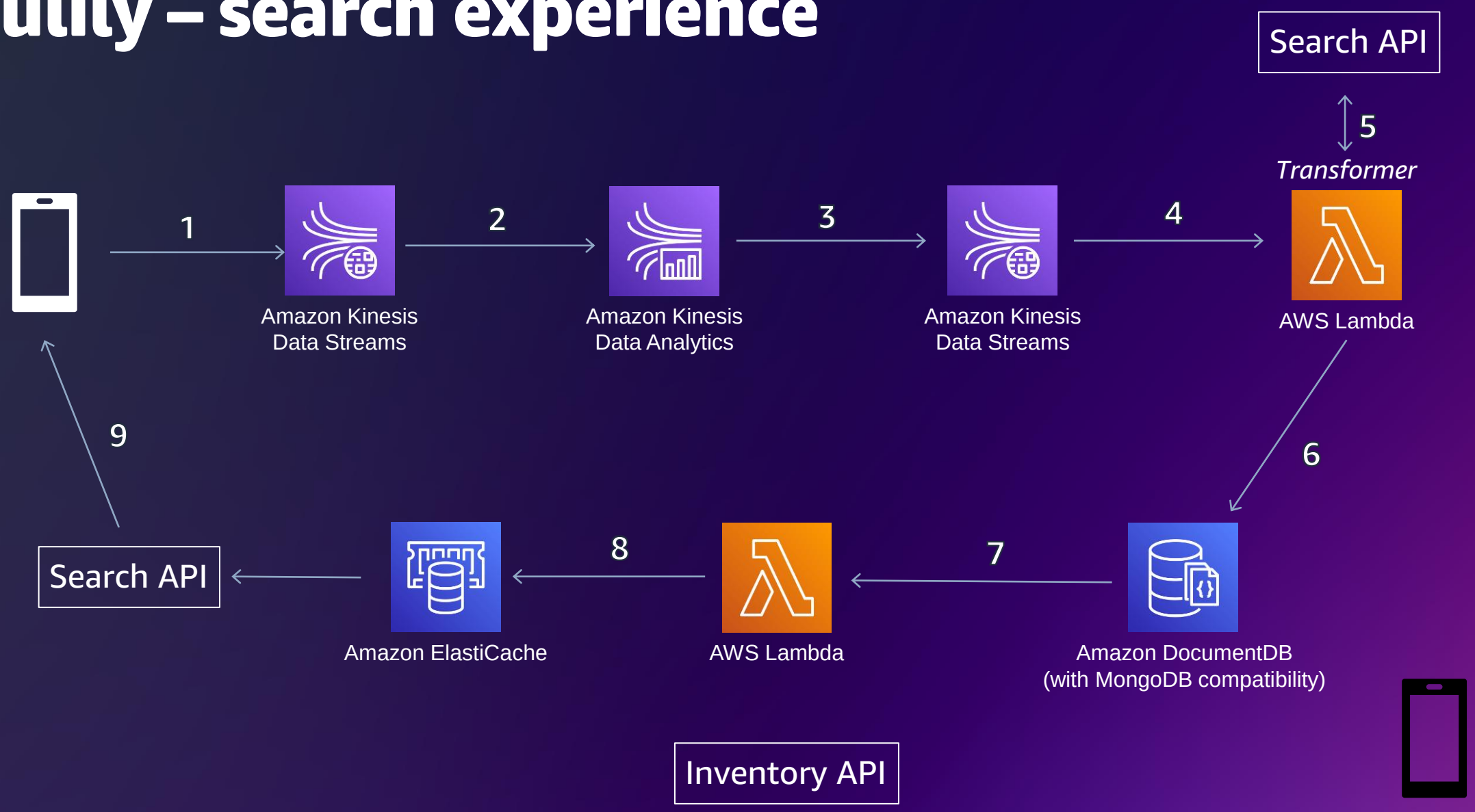
Zulily – search experience



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Wide column: Apache Cassandra

Open-source, wide-column data store

Cassandra Query Language (CQL)

Large-scale applications that require fast read and write performance

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Amazon Keyspaces (for Apache Cassandra)

Scalable, highly available, and managed Apache Cassandra-compatible database service

Apache Cassandra-compatible

Use the same Cassandra drivers and tools

No servers to manage

No need to provision, configure, and operate large Cassandra clusters

Single-digit millisecond performance at scale

Scale tables up and down automatically

Virtually unlimited throughput and storage

Highly available and secure

99.99% availability SLA within an AWS Region

Data encrypted at rest; integrated with IAM

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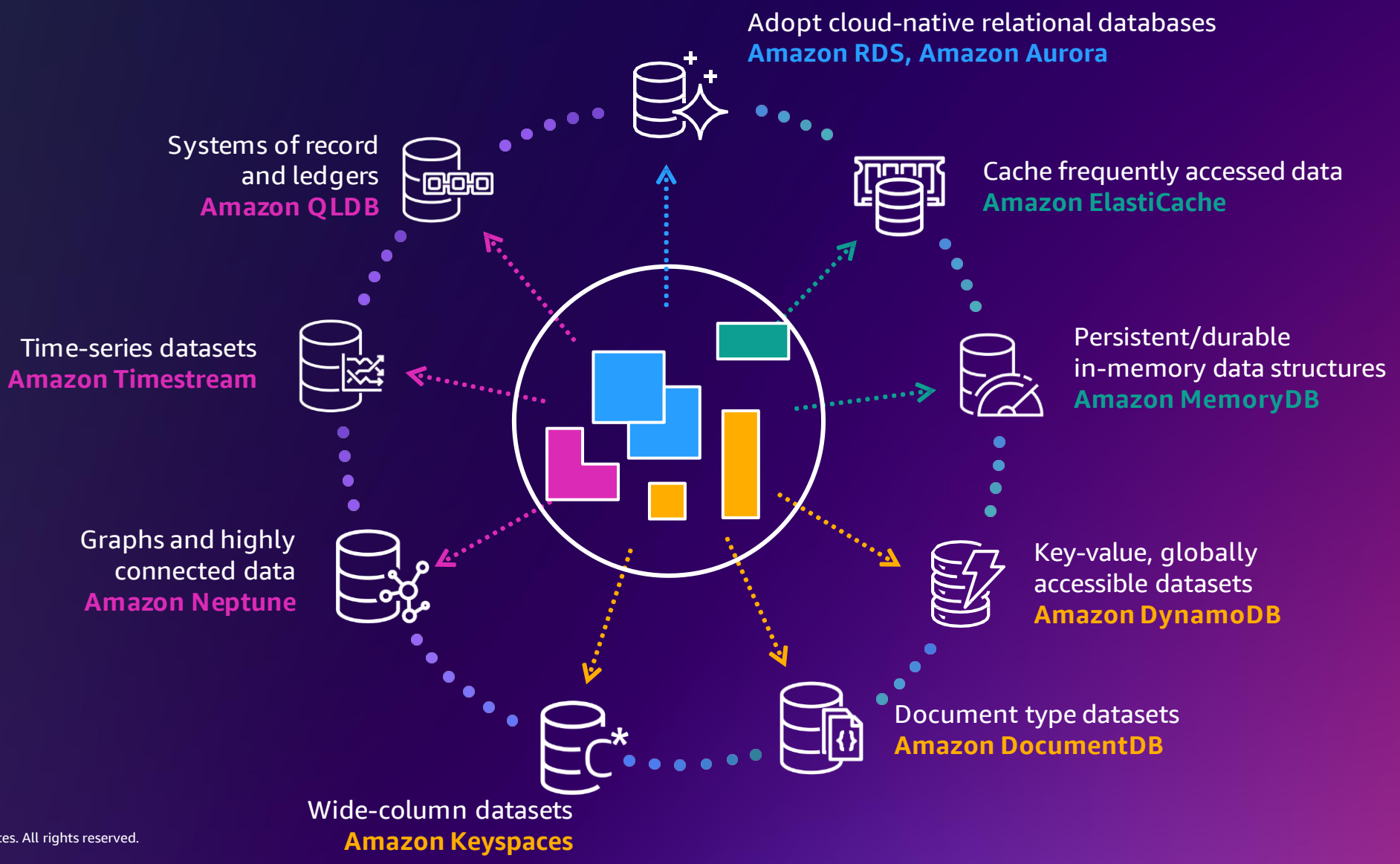
Step 4: Specialized databases

Specialized datasets

- Social graphs
- Recommendation engines
- Measurements
- Systems of record
- Digital records

Benefits

- Agility
- Scalability
- Performance



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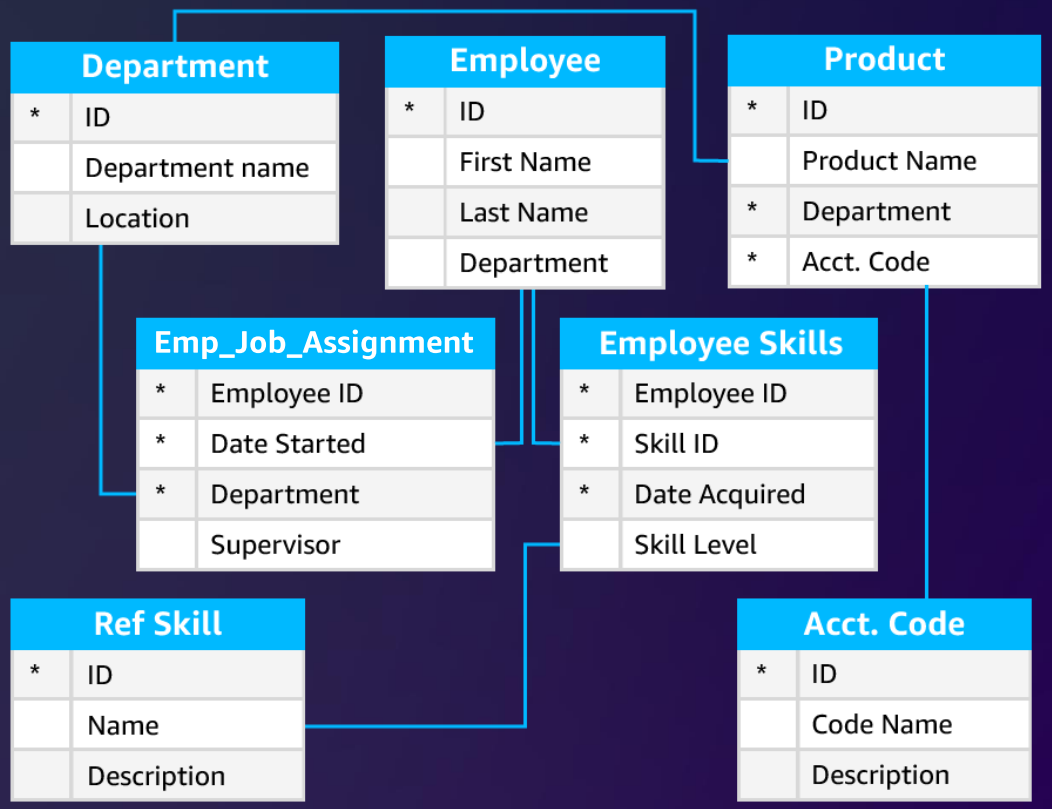
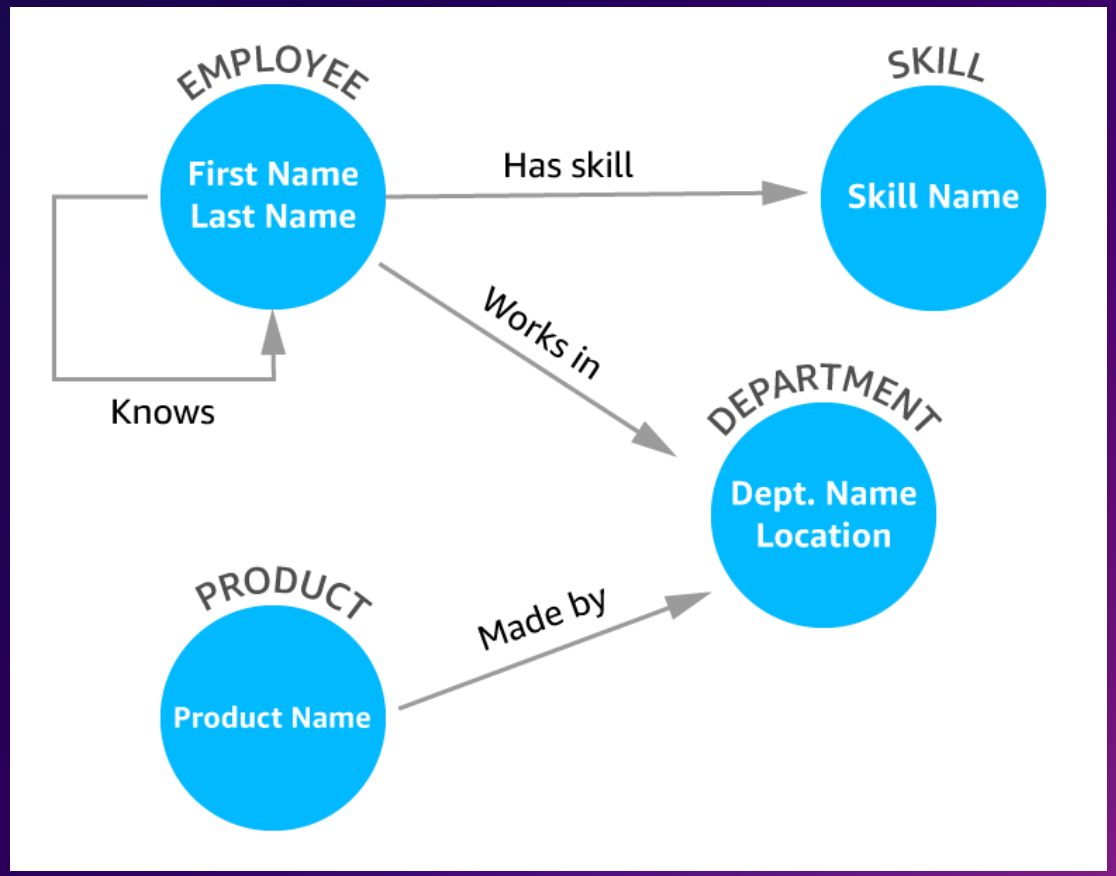
Different approaches for highly connected data



Purpose-built for a business process



Purpose-built to answer questions about relationships

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Amazon Neptune

Fast, reliable graph database built for the cloud

Open

Supports Apache TinkerPop & W3C RDF graph models

Fast

Query billions of relationships with millisecond latency

Reliable

6 replicas of your data across 3 AZs with full backup and restore

Easy

Build powerful queries easily with Gremlin, SPARQL and openCypher

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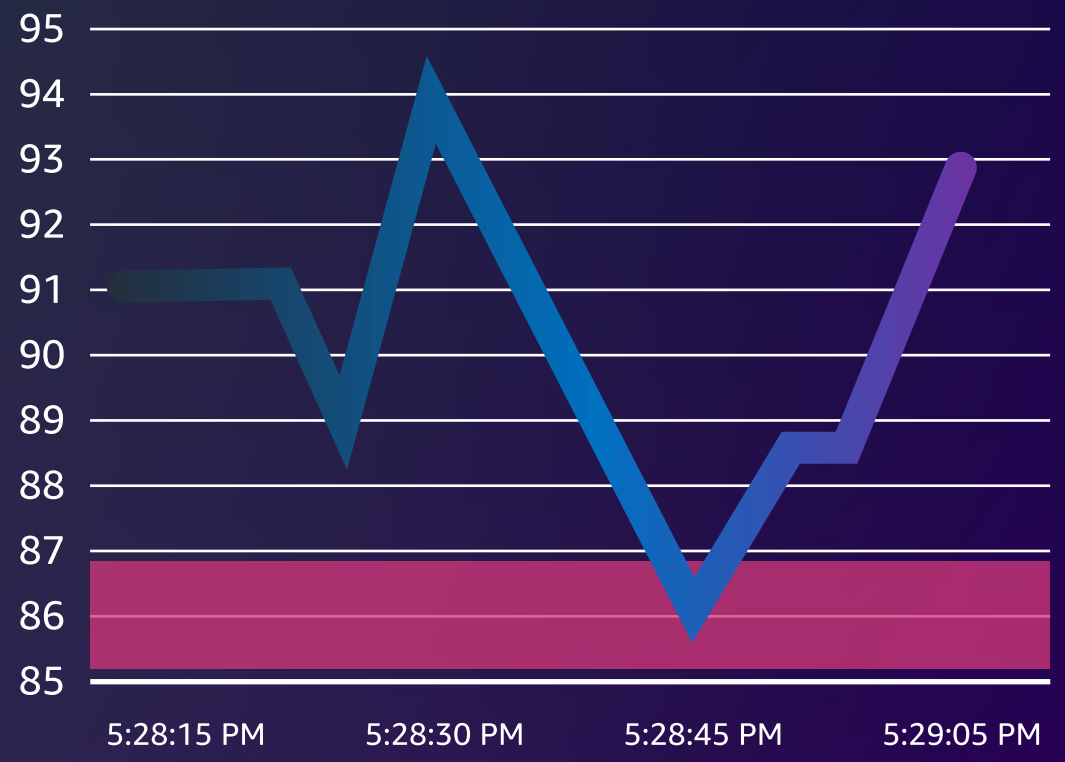
Time-series use cases



Humidity

91.0

% water vapor

1. Application events
2. IoT sensor readings
3. DevOps data

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Amazon Timestream

Fast, scalable, and fully managed time-series database

1,000x faster and 1/10th the cost of relational databases

Collect data at the rate of millions of inserts per second (10M/second)

Trillions of daily events

Adaptive query processing engine maintains steady, predictable performance

Time-series analytics

Built-in functions for interpolation, smoothing, and approximation

Serverless

Automated setup, configuration, server provisioning, and software patching

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Ledgers: immutable and verifiable data

Risk reduction

Ensure safeguarding of critical system-of-record applications where data loss could be expensive

Data tracking improvements

Track data's entire lineage quickly and accurately, improving efficiency in identifying source of issues

Auditing and compliance

Reduce downtime caused by audit and compliance issues

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Amazon Quantum Ledger Database (QLDB)

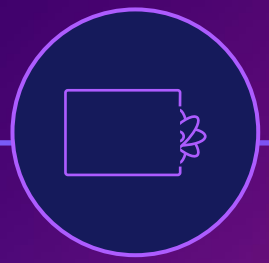
Amazon QLDB is a fully managed ledger database that maintains a complete, immutable, and verifiable history of all changes over time.

Serverless

Document-oriented

Append only

Cryptographic hashing



Fully serialized

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Purpose-built modernization is a journey

BOTH ORGANIZATIONALLY AND TECHNOLOGICALLY

1. **Move relational workloads to managed DB services**
   Aurora, Amazon RDS
   Improve: agility, ease of management, cost-efficiency

2. **Implement caching and offload in-memory workloads**
   ElastiCache, MemoryDB
   Improve: scalability and performance

3. **Move nonrelational query patterns to NoSQL**
   DynamoDB, Amazon DocumentDB, Amazon Keyspaces
   Improve: agility, innovation, scalability, performance, ease of management, cost-efficiency

4. **Move specialized datasets and query patterns to specialized databases**
   Neptune, Timestream, Amazon QLDB
   Improve: agility, innovation, scalability, performance, ease of management, cost-efficiency

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Thank you!

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